

太湖学院机械 65 班《大学物理 (1)》A 卷 答案 2007,6

本题
得分

一、选择题(每题 2 分,共 20 分)

题号	1	2	3	4	5	6	7	8	9	10
答案	C	B	A	C	D	B	C	B	D	C

本题
得分

二、填空题(每题 2 分,共 20 分)

题号	1	2	3	4	5
答案	20 m/s	$\sqrt{\frac{3g}{l}}$	12 J	$\frac{q}{4\pi\epsilon_0 a(a+L)}$	$160\text{ N}\cdot\text{s}$
题号	6	7	8	9	10
答案	2	$\frac{\mu_0 I l}{2\pi} \ln \frac{b}{a}$	$\frac{q}{6\epsilon_0}$	$\frac{qQ}{4\pi\epsilon_0 R}$	$-\sigma$

三、(本题 10 分)

$$mg - T = ma \quad (2\text{分}) \quad a = \frac{mg}{m + \frac{M}{2}} = \frac{1}{3}g \quad (3\text{分})$$

$$TR = J\alpha = \frac{1}{2}MR^2 \cdot \frac{a}{R} \quad (3\text{分}) \quad T = \frac{2}{3}mg \quad (2\text{分})$$

四、(本题 10 分)

$$F = m \frac{dv}{dt} = m \frac{dv}{dx} \cdot \frac{dx}{dt} = mv \frac{dv}{dx} = 3 + 4x \dots (4\text{分})$$

$$\int_0^v v dv = \int_0^x (3 + 4x) dx, \dots (2\text{分})$$

$$\frac{1}{2}v^2 = 3x + 2x^2 \dots (2\text{分})$$

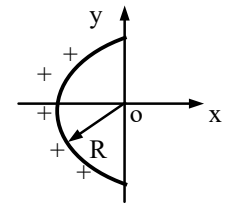
$$v = \sqrt{2(3x + 2x^2)} \dots (2\text{分})$$

五、(本题 10 分)

$$dq = \frac{qRd\theta}{\pi R} = \frac{q}{\pi} d\theta, (1\text{分}), dE = \frac{dq}{4\pi\epsilon_0 R^2} = \frac{qd\theta}{4\pi^2\epsilon_0 R^2} (2\text{分})$$

$$dE_x = dE \cdot \sin \theta, dE_y = -dE \cdot \cos \theta (2\text{分}), \int dE_y = 0 (1\text{分})$$

$$\therefore E = E_x = \int_0^\pi dE_x = \frac{q}{2\pi^2\epsilon_0 R^2} (3\text{分}), \text{方向指向}x\text{轴正向} (1\text{分})$$

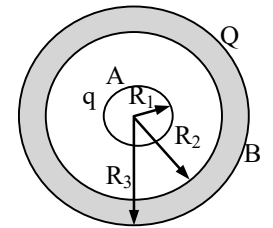


六、(本题 10 分)

$$(1) 1, r < R_1, E_1 = 0 \dots (2\text{分}) \quad 2, R_1 < r < R_2, E_2 = \frac{q}{4\pi\epsilon_0 r^2} \dots (2\text{分})$$

$$3, R_2 < r < R_3, E_3 = 0 \dots (2\text{分}) \quad 4, r > R_3, E_4 = \frac{q+Q}{4\pi\epsilon_0 r^2} \dots (2\text{分})$$

$$(2) U_{AB} = \int_{R_1}^{R_2} \vec{E}_2 \cdot d\vec{l} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \dots (2\text{分})$$



七、(本题 10 分)

$$(1) r < R_1, \oint \vec{B} \cdot d\vec{l} = \mu_0 \sum I_i, B_1 = \frac{\mu_0 I r}{2\pi R^2} (4\text{分})$$

$$(2) R_1 < r < R_2, B_2 = \frac{\mu_0 I}{2\pi r} (3\text{分})$$

$$(3) r > R_2, B_3 = 0 (3\text{分})$$

八、(本题 10 分)

$$AO: B_1 = \frac{\mu_0 I}{4\pi r} (\cos 0^\circ - \cos 150^\circ) = \frac{\mu_0 I}{4\pi r} \left(1 + \frac{\sqrt{3}}{2} \right) (4\text{分})$$

$$OC: B_2 = \frac{\mu_0 I}{4\pi r} (\cos 150^\circ - \cos 180^\circ) = \frac{\mu_0 I}{4\pi r} \left(1 - \frac{\sqrt{3}}{2} \right) (4\text{分})$$

$$B_P = B_1 + B_2 = \frac{\sqrt{3}\mu_0 I}{4\pi r} (2\text{分})$$